2015 EXAMINATIONS

PART I

ECONOMICS

ECON 102 Principles of Economics **(3 hours)**

Candidates should answer ALL questions from Section A, ONE question from Section B and ONE question from Section C.

Section A (40 marks)
Identify the correct answers on the answer sheet provided.

1. The scarcity principle indicates that
   a. No matter how much one has, it is never enough.
   b. Compared to 100 years ago, individuals have less time today.
   c. With limited resources, having more of one good or service means having less of another.
   d. Because tradeoffs must be made, resources are scarce.

2. One observes that the equilibrium price of T-shirts increases and the equilibrium quantity falls. Which of the following best fits the observed data?
   a. An increase in demand with supply constant.
   b. A decrease in supply with demand constant.
   c. An increase in demand coupled with an increase in supply
   d. A decrease in demand with supply constant

3. You have one hour to catch a flight to Madrid for spring break. It takes 45 minutes to drive to the airport. Your car is almost out of gas; the price of gas at the closest gas station is higher than the one other side of the town. To you, the price elasticity of demand for gas is likely to be _____ than it would be if you had several hours before the flight.
   a. More inelastic
   b. More elastic
   c. Unit elastic
   d. No different

Please turn over

4. Which of the following will NOT cause a shift in the demand curve for iPhones?
   a. An increase in the price of iPhone apps.
   b. A decrease in the price of the Windows phone.
   c. A decrease in the price of iPhones.
   d. An increase in consumers' incomes.

5. At his current consumption level Cameron get 3 times more marginal utility from an additional game of pinball than from an additional game of Spiderman. If the price of Spiderman games is £0.50, then Cameron is maximizing utility if the price of a pinball game is:
   a. £1.00
   b. £1.50
   c. £2.00
   d. £3.00

6. Which of the following is NOT true of a perfectly competitive firm?
   a. It faces a perfectly elastic demand curve.
   b. It is unable to influence the market price of the good it sells.
   c. Relative to the size of the market, the firm is small.
   d. It seeks to maximize revenue.

7. Economic profit is:
   a. Always greater than or equal to accounting profit.
   b. Always less than or equal to accounting profit.
   c. Always equal to accounting profit.
   d. Only important to economists.

8. For perfectly competitive firms price _____ marginal revenue, and for monopolists price _____ marginal revenue.
   a. Equals; is greater than.
   b. Equals; equals
   c. Equals; is less than
   d. Is greater than; equals

9. A monopolist faces the demand curve: $P(Q) = 21 – 2Q$, and marginal cost curve: $MC(Q) = 3Q$. The profit-maximizing quantity is_____, and the profit-maximizing price is _____.
   a. Less than 5; less than 10.
   b. Less than 5; greater than 10.
   c. Greater than 5; greater than 10.
   d. Greater than 5; less than 10.

10. Stefanie is looking to sell her Playstation 4; she has a reservation price of £300. Dave is looking to buy a Playstation 4; he has a reservation price of £500. If a transaction takes place and Dave buys Stefanie's Playstation 4, the total amount of economic surplus generated between Dave and Stefanie:
    a. Depends on the price Dave pays Stefanie for the Playstation 4.
    b. Is equal to £300.
    c. Is equal to £500.
    d. Is equal to £200.

11. A Nash equilibrium is:
    a. A strategy that yields a higher payoff, no matter what the other players in a game choose.
    b. A strategy that yields a lower payoff, no matter what the other players in a game choose.
    c. Any combination of strategies in which each player's strategy is his best choice, given the other player's strategies.
    d. None of the above.

12. In the ultimatum bargaining game:
    a. The second mover can make a credible threat.
    b. The second mover gets most of the payoff.
    c. The first mover gets most of the payoff.
    d. Both players move simultaneously.

13. In a Cournot model of duopoly:
    a. Total profits of the two firms is greater than the monopoly profit.
    b. Total output of the two firms is lower than the monopoly output.
    c. The reaction function of Firm 1 is flatter than the reaction function of Firm 2, when the quantity of firm 1 is on the horizontal axis.
    d. The reaction function of Firm 1 is steeper than the reaction function of Firm 2, when the quantity of firm 1 is on the horizontal axis.

14. A monopsony:
    a. Is a single seller of a good.
    b. Will pay a higher wage than the competitive wage.
    c. Sets a higher wage with a trade union than without a trade union.
    d. Will employ more workers than a competitive labour market.

15. A minimum wage:
    a. Always increases total economic surplus.
    b. Always increases employer surplus.
    c. Always increases worker surplus.
    d. None of the above.

16. A country has a comparative advantage in a good if:
    a. A country can produce a good using fewer factor inputs than another country.
    b. A country can produce a good using more factor inputs than another country.
    c. A country has a lower opportunity cost than another country.
    d. A country has a higher opportunity cost than another country.

17. When a country imposes an import tariff:
    a. Consumers are better off.
    b. Producers are worse off.
    c. The government obtains revenue from the tariff.
    d. The country overall is better off.

18. With an externality:
    a. The level of the activity will always be less than the socially optimal level.
    b. The level of the activity will always be more than the socially optimal level.
    c. The level of the activity will always be equal to the socially optimal level.
    d. None of the above.

19. The tragedy of the commons:
    a. Is the problem of unpriced resources.
    b. Is a positive externality.
    c. Cannot be solved through private ownership of the commons.
    d. Cannot be solved through collective decision-making.

20. Moral hazard is:
    a. When insurance is purchased disproportionately by those who are most costly for companies to insure.
    b. The outcome of a market for lemons.
    c. The tendency of people to expend less effort protecting those goods that are insured against theft or damage.
    d. Something that cannot be overcome.

21. In a certain fictional country the value of Output (in billions) is $1,600, the population number is 80 million and the number of people employed is 40 million. What is the output per person and the average labour productivity in this country?
    a. Output per person=$20,000; Average labour productivity=$40,000 per person.
    b. Output per person=$40,000; Average labour productivity=$20,000 per person.
    c. Output per person=$20; Average labour productivity=$40 per person.
    d. Output per person=$30,000; Average labour productivity=$50,000 per person.

22. In Panem state, the expenditure as a percentage of GDP is 50% and the revenue as a percentage of GDP is 35%. What is the value of the government deficit / surplus in this state?
    a. Surplus of 15 percent out of GDP.
    b. Surplus of 85 percent out of GDP.
    c. Deficit of 15 percent out of GDP.
    d. Deficit of 85 percent out of GDP.

23. Assume a country which produces only Apples and Pears. In a given year this country produces 1,000 apples at the price of $2 per apple, and 500 pears at the price of $1 per pear. What is the aggregate output for this country?
    a. 2000.
    b. 500.
    c. 3000.
    d. 2500.

24. During 2014, Katniss earned a nominal annual salary of $52,500. If the CPI was 105 during 2014, what was Katniss real income during this period?
    a. 50,000.
    b. 51,500.
    c. 52,000.
    d. 52,500.

25. A fictional town has a labour force of 1,500, with 100 people being unemployed. What is the unemployment rate in this town?
    a. 0.55%.
    b. 5.05%.
    c. 6.66%.
    d. 10.0%.

26. The demand equation for employment is given by N_d =400-2w whereas the supply of labour is given by N_s =240+2w. The terms N_d, N_s and w represent labour demand, labour supply and wages, respectively. What is the equilibrium level of employment and wages in this economy?
    a. w=40 and N=320.
    b. w=30 and N=500.
    c. w=25 and N=250.
    d. None of the above.

27. Assume a poor economy gets a large inflow of investment which increases its capital stock. The production curve (as a function of capital and labour):
    a. goes up.
    b. goes down.
    c. remains the same but with a higher capital-labour ratio.
    d. it is impossible to know.

28. A certain bond is expected to pay a dividend of $10 in one year and sell for $100 at that time. If the safe return on this bond is 5% in addition to a risk premium of 3%, how much would you be willing to pay today for one unit of this bond?
    a. $95.25.
    b. $101.85.
    c. $99.55.
    d. the bond carries risk and therefore I will not be willing to pay for it.

29. Assume consumption expenditures=1000, investment=2000, government purchases=500. What is the gross domestic product (Y) and the national saving rate (S/Y)?
    a. Y=3500, S/Y=57.14%.
    b. Y=3500, S/Y=60.00%.
    c. Y=3000, S/Y=55.54%.
    d. none of the above.

30. For an economy with a consumption function of: C=0.80(Y-T), where Y denotes output and T denotes taxes, what is the value of the marginal propensity to consume (MPC) and the income-expenditure multiplier (IEM) .
    a. MPC=0.80, IEM=4.
    b. MPC=0.80, IEM=3.
    c. MPC=0.75, IEM=5.
    d. MPC=0.80, IEM=5.

Please turn over

31. For an economy characterized by: C=1800+0.60(Y-T), I=900, G=1,500, NX=100, T=1,500, what is the equilibrium level of output (Y)?
    a. 8000.
    b. 8500.
    c. 9000.
    d. 9500.

32. What policy should the central bank pursue if it wishes to prevent an increase in the equilibrium interest rate following a rise in the price level?
    a. Increase the money supply.
    b. Reduce the money supply.
    c. Keep the money supply unchanged.
    d. Central bank has no power to control the equilibrium interest rate.

33. In the IS-LM model, an increase in government expenditures will:
    a. Shift the IS curve to the right (upwards).
    b. Increase the equilibrium interest rate.
    c. Reduce the level of investments.
    d. All of the above.

34. If the central bank reduces the money supply:
    a. The LM curve shifts to the right.
    b. The equilibrium interest rate goes up.
    c. Output goes up.
    d. None of the above.

35. In a liquidity trap:
    a. The money demand function is perfectly elastic with respect to interest rate changes.
    b. Fiscal policy is ineffective.
    c. Monetary policy is very effective.
    d. None of the above.

36. Within the Keynesian model, the aggregate supply schedule determines whether an increase in aggregate demand is likely to
    a. lead to a fiscal deficit
    b. lead to monetary expansion
    c. lead to exchange rate depreciation
    d. none of the above

Please turn over

37. ‘A year ago, QE might have stopped the appreciation of the euro, one of the factors behind the fall in (eurozone) inflation’ (*Financial Times*, 19/5/2014). QE might have stopped the appreciation of the euro, because QE
    a. monetizes sovereign debt
    b. raises interest rates
    c. attracts short-term capital into the eurozone
    d. all of the above

38. If job-seekers under-estimate the rate of inflation, the duration of job-search tends to
    a. fall, so that unemployment tends to rise
    b. rise, so that unemployment tends to fall
    c. fall, so that unemployment tends to fall
    d. rise, so that unemployment tends to rise

39. Monetarism assumes a close and stable relationship between
    a. real balances and the transactions demand for money
    b. inflation and unemployment
    c. nominal money supply and nominal income
    d. wages and prices

40. For UK international payments, the ‘balance for official financing’ is the value of
    a. foreign exchange bought/sold by the authorities to set the value of sterling
    b. net exports including ‘invisibles’
    c. foreign exchange reserves
    d. net UK borrowing from foreign central banks

Section B (30 marks) Answer ONE question

41. Bundles of wheat are sold in a perfectly competitive market. The supply curve for wheat is given by: p = 20 + 4Q. The demand curve for wheat is given by: p = 92 – 2Q. The price is given in £, and the quantity is bundles of wheat. Each part of this question is weighted equally.

a. Calculate the equilibrium price and quantity of wheat, consumer and producer surplus.

b. Draw a graph representing this market. On your graph, label the equilibrium price and quantity of wheat, and show consumer and producer surplus.

Suppose the government passes legislation, which provides a £12 per-bundle subsidy to sellers of wheat.

c. Calculate the equilibrium price and quantity of wheat, consumer surplus, and producer surplus under the subsidy.

d. Calculate the cost to the government and the deadweight loss under the subsidy.

e. Imagine you work for a policy maker, who has hired you to evaluate whether or not the wheat subsidy should be abandoned. In 50 words or less, provide a recommendation to the policy maker, and support your case.

42. Answer all of the following parts of this question.

a. The demand function of a firm is given by $\log_e q = -3p + 2\log_e y$ where q is the quantity demanded, p is price and y is real income.

Determine the price elasticity of demand. In what range of p is the price elasticity of demand inelastic? (8%)

b. The utility, U, of an individual i is given by $U = C^{0.8}l^{0.2}$ where C is real consumption and l is leisure.

Derive the first and second derivatives of utility with respect to consumption, as well as the cross partial derivative of utility with respect to consumption and leisure. What is the economic interpretation of these derivatives? (8%)

c. A firm's production function is given by $q = 10L + 10K + 2KL - 4L^2 - 1.5K^2$ where q is real output, L is the labour input and K is the capital input. What values of L and K will maximise the firm's output level? (9%)

d. $y = 10x^{-1}z^{-2}a^{-3} - 5\log_e x + 4e^z - x\log_e z$
Determine (i) $\frac{\partial y}{\partial a}$ (ii) $\frac{\partial y}{\partial x}$ (iii) $\frac{\partial y}{\partial z}$ (5%)

43. Answer all of the following parts of this question.

a. What is an externality? Draw a diagram showing the impact of an externality on the output of a good.

b. What is a public good? Draw a diagram showing the demand curve of a public good and how the output of a public good is determined.

Section C (30 marks) Answer ONE question

44. Consider the following economy:

$$C=200+0.75(Y-T)$$
$$I=200-25r$$
$$T=100,\ G=100$$
$$(M/P)^D=Y-100r$$
$$P=2.\ M^s=1000$$

C=consumption, Y=output, I=investment, T=taxes, G=government spending, r=interest rate, $(M/P)^D$=real demand for money, P=prices and M^s=money supply.

a. Find the algebraic expression for the IS curve and the LM curve and explain why the slope of the IS curve is negative while the slope of the LM curve is positive.

b. Find the equilibrium interest rate (r) and the equilibrium level of output (Y).

c. Suppose the government increases its purchases from 100 to 150. What is the new equilibrium level of interest rate and output? Explain your answer using words and a relevant graph.

45. Define national saving, relating your definition to the general concept of saving.

46. By the theoretical specification of expectations-augmented Phillips curves, explain how these curves stack 'upwards', *not* 'outwards'. Explain the relevance of this analysis to the 'Heath-Barber Boom', 1972-74.

End of paper